Winding Extensions of the Kari-Culik Shift

Cyclic covers, relative eigenfields, and a nonconjugacy program

Referee draft, 27 July 2026

Abstract

For each integer $m \geq 1$, we define a cyclic winding extension $K^{(m)}$ of the 13-tile Kari-Culik Wang shift. The sheet coordinate is constant along each row and advances by one precisely across a division row. The extension is an m-to-one finite cover of the Kari-Culik shift. It is strongly aperiodic, and it has the same positive topological entropy as its base.

Assuming the Siefken subsystem input stated below, balanced discrepancy on its lift gives a continuous transfer function and an explicit vertical eigenfunction with eigenvalue

$$\exp\left(2\pi i \frac{\theta}{m}\right), \qquad \theta = \frac{\log 2}{\log 6}.$$

We also give an abstract construction that turns every fixed-frame eigenfunction of a geometric suspension into a relative eigenfield on its complete rotating hull.

A proposed computation of the full fixed-frame eigenmodule would distinguish the winding indices m. Several steps needed for that computation remain unproved: a controlled return lemma in Siefken's parameter space, exclusion of horizontal eigenvalues, analysis of the finite phase extension, and completeness of the resulting module. The pairwise nonconjugacy statement is therefore stated conditionally.

1. Scope and status

The winding construction arose as the dynamical component of a geometric compiler from Wang shifts to polygonal tiling hulls. Its symbolic content is independent enough to merit separate treatment. This paper isolates that content and distinguishes established statements from the arguments that still require specialist review.

The status of the main statements is as follows.

Statement	Status in this draft
$K^{(m)} \to K$ is an m-to-one cyclic cover	Proved
$K^{(m)}$ is strongly aperiodic	Proved from the aperiodicity of K
$h_{\mathrm{top}}(K^{(m)}) = h_{\mathrm{top}}(K) > 0$	Proved
$e(\theta/m)$ is a vertical eigenvalue on the lifted Sturmian subsystem	Proved conditional on the stated Siefken subsystem input; exact citation pending
The H-invariant vertical eigenvalue group is generated by $e(\theta/m)$	Conditional
The lifted subsystem has no nontrivial horizontal eigenvalue	Open
The four-phase extension has the displayed full eigenvalue group	Conditional
The displayed fixed-frame module M_m is complete	Open
Rotating Siefken-component hulls with distinct winding indices are nonconjugate	Conditional
Full compiler hulls with distinct winding indices are nonconjugate	Open; requires intrinsic component recognition

Here and below,

$$e(t) = \exp(2\pi i t).$$

2. The Kari-Culik shift

2.1 The thirteen tiles

Write a Wang tile as (S, W, N, E). The six division tiles are

S	W	N	E
1	0/3	0	1/3
1	1/3	0	2/3
1	2/3	1	0/3
2	0/3	0	2/3
2	1/3	1	0/3
2	2/3	1	1/3

The seven doubling tiles are

S	W	N	E
0	-1	$0'$	-1
0	0	$0'$	0
0	0	1	-1
1	-1	1	0
1	-1	2	-1
1	0	2	0
$0'$	0	1	-1

The colors 0 and $0'$ are distinct. The expressions $0/3, 1/3, 2/3$ are formal colors. Let K be the nearest-neighbor $\mathbb{Z}^2$-shift defined by matching equal colors on shared edges. Let H and V denote

the horizontal and vertical shifts.

Define the multiplier type

$$c(\tau) = \begin{cases} 1, & \tau \text{ is a division tile,} \\ 0, & \tau \text{ is a doubling tile.} \end{cases}$$

Lemma 2.1, row type. In every $x \in K$, the value $c(x_{i,j})$ is independent of i. Thus each row has a well-defined multiplier type $c_j(x) \in \{0, 1\}$.

Proof. Every west and east color of a division tile belongs to $\{0/3, 1/3, 2/3\}$. Every west and east color of a doubling tile belongs to $\{-1, 0\}$. These two sets of formal colors are disjoint. Horizontal matching therefore prevents a division tile and a doubling tile from being adjacent. Since a row is connected, its type is constant. □

We use two established properties of K:

1. K is nonempty and has no periodic configuration, by Culik's construction and aperiodicity theorem.
2. K has positive topological entropy, by Durand, Gamard, and Grandjean.

The eigenvalue discussion will use Siefken's distinguished minimal subsystem and its Sturmian parameterization.

3. The cyclic winding extensions

3.1 Definition

Fix $m \geq 1$. Replace each Kari-Culik tile τ by the m symbols

$$(\tau, r), \qquad r \in \mathbb{Z}/m\mathbb{Z},$$

with edge colors

$$S(\tau, r) = (S_\tau, r), \quad W(\tau, r) = (W_\tau, r),$$
$$E(\tau, r) = (E_\tau, r), \quad N(\tau, r) = (N_\tau, r + c(\tau)).$$

Let $K^{(m)}$ be the Wang shift defined by these decorated tiles, and let

$$\pi_m : K^{(m)} \longrightarrow K, \qquad \pi_m(\tau, r) = \tau$$

be the coordinatewise projection.

Horizontal matching forces the sheet coordinate to be constant along a row. If r_j is its value on row j, vertical matching gives

$$r_{j+1} = r_j + c_j \pmod{m}. \tag{3.1}$$

Proposition 3.1, finite-cover structure. The map π_m is onto, and every $x \in K$ has exactly m lifts to $K^{(m)}$.

Proof. By Lemma 2.1, each row of x has a type c_j. Choose $r_0 \in \mathbb{Z}/m\mathbb{Z}$. Equation (3.1) determines r_j uniquely for all positive and negative j. Assigning r_j to every tile of row j satisfies all horizontal and vertical decorated-edge constraints. This gives a lift.

Conversely, horizontal matching makes the sheet coordinate constant on each row of any lift, and vertical matching forces (3.1). A lift is therefore determined by r_0. The m choices of r_0 give distinct lifts. □

3.2 Strong aperiodicity

We use the following standard reduction.

Lemma 3.2, one period produces two. Let X be a two-dimensional shift of finite type. If X contains a point with a nonzero translational period, then X contains a doubly periodic point.

Proof. Let $p \in \mathbb{Z}^2 \setminus \{0\}$ be a period of $x \in X$. Configurations invariant under p can be represented on the finite-width cylinder $\mathbb{Z}^2/\langle p \rangle$. Reading this cylinder in any integral direction independent of p produces a one-dimensional shift of finite type with a finite alphabet of cylinder slices. It is nonempty because it contains the image of x. Every nonempty one-dimensional shift of finite type contains a periodic point, obtained from a directed cycle in a finite presentation. Lifting that point to $\mathbb{Z}^2$ gives a configuration with two independent periods. □

Corollary 3.3. Every configuration in K has no nonzero period.

Proof. A nonzero period would give a doubly periodic point by Lemma 3.2, contrary to the aperiodicity theorem for the Kari-Culik tile set. □

Theorem 3.4. For every $m \geq 1$, the shift $K^{(m)}$ is nonempty and strongly aperiodic.

Proof. Nonemptiness follows from Proposition 3.1. Any period of a point in $K^{(m)}$ would project under π_m to the same period of a point in K, contradicting Corollary 3.3. □

3.3 Entropy

Let $p_X(N)$ denote the number of globally admissible $N \times N$ patterns in a shift X.

Proposition 3.5. For every $N \geq 1$,

$$p_{K^{(m)}}(N) = m\, p_K(N). \tag{3.2}$$

Consequently,

$$h_{\mathrm{top}}(K^{(m)}) = h_{\mathrm{top}}(K) > 0.$$

Proof. A globally admissible base pattern occurs in some $x \in K$. For each of the m choices of sheet on one row, the recurrence (3.1) gives a decorated pattern, and Proposition 3.1 extends it globally. These decorated patterns are distinct.

Every decorated $N \times N$ pattern projects to a base pattern. Its sheet values are constant on rows and determined by one initial value, so the fiber over a base pattern has at most m elements. This proves (3.2). After taking logarithms, dividing by N^2, and passing to the limit, the factor m disappears. Positivity is the theorem of Durand, Gamard, and Grandjean. □

4. A Sturmian cocycle and an explicit eigenvalue

4.1 The multiplier factor

Set

$$\theta = \frac{\log 2}{\log 6}.$$

This number is irrational. Indeed, a relation $\log 2/\log 6 = p/q \in \mathbb{Q}$ would give $2^q = 6^p$, which is impossible for positive integers p, q by unique factorization.

We isolate the following published input attributed to Siefken. Its precise location and formulation in the cited paper must be supplied before submission.

Subsystem input 4.1. There is a minimal Kari-Culik subsystem $Y \subseteq K$ with a continuous parameter model such that the multiplier itinerary

$$c : Y \longrightarrow \{0,1\}^{\mathbb{Z}}, \qquad c(y)_j = c_j(y),$$

is invariant under H and intertwines V with the shift σ. On this subsystem its image is the Sturmian shift C_θ of slope θ. The parameter model also supplies row parameters α_j whose logarithmic coordinates

$$u_j = \frac{\log \alpha_j + \log 3}{\log 6}$$

satisfy

$$u_{j+1} = u_j + \theta - c_j \pmod 1. \tag{4.1}$$

Equation (4.1) explains the occurrence of the rotation number θ. The arguments below are restricted to the Sturmian factor C_θ. They make no claim about multiplier itineraries in every component of the full Kari-Culik hull.

4.2 The transfer function

Lemma 4.2. Assuming Subsystem input 4.1, there is a continuous function

$$h : C_\theta \longrightarrow \mathbb{R}$$

such that

$$h(\sigma c) - h(c) = c_0 - \theta \tag{4.2}$$

for every $c \in C_\theta$.

Proof. The Sturmian shift is minimal. Its words are balanced, so the Birkhoff sums

$$\sum_{j=0}^{N-1} (c_j - \theta)$$

are uniformly bounded over $c \in C_\theta$ and $N \geq 1$. The Gottschalk-Hedlund theorem applied to the continuous cylinder function $c \mapsto c_0 - \theta$ supplies a continuous transfer function satisfying (4.2). $\square$

The transfer function is unique up to an additive constant because C_θ is minimal.

4.3 The lifted eigenfunction

Let

$$Y^{(m)} = \pi_m^{-1}(Y) \subseteq K^{(m)}.$$

This finite extension need not itself be minimal. All eigenfunctions below are defined on all of $Y^{(m)}$, and by convention they are circle-valued. For $z \in Y^{(m)}$, write $r(z)$ for the sheet coordinate at the origin and continue to write $c(z)$ for the multiplier itinerary of its projection.

We choose the shift convention in which

$$\begin{aligned} r(Hz) &= r(z), & c(Hz) &= c(z), \\ r(Vz) &= r(z) + c(z)_0, & c(Vz) &= \sigma c(z). \end{aligned} \tag{4.3}$$

For (4.4), any integer representative of $r(z) \in \mathbb{Z}/m\mathbb{Z}$ may be used. Adding m to the representative leaves the value unchanged.

Theorem 4.3, a vertical eigenvalue. Assuming Subsystem input 4.1, the function

$$F_m(z) = e\left(\frac{r(z) - h(c(z))}{m}\right) \tag{4.4}$$

is continuous and satisfies

$$F_m(Hz) = F_m(z), \qquad F_m(Vz) = e(\theta/m)F_m(z). \tag{4.5}$$

Proof. Continuity follows because $r(z)$ is a cylinder coordinate, c is a continuous factor map, and h is continuous. Horizontal invariance follows from (4.3). For the vertical action, equations (4.2) and (4.3) give

$$\begin{aligned} F_m(Vz) &= e\left(\frac{r(z) + c(z)_0 - h(\sigma c(z))}{m}\right) \\ &= e\left(\frac{r(z) - h(c(z)) + \theta}{m}\right) \\ &= e(\theta/m)F_m(z). \end{aligned}$$

□

A continuous joint eigenfunction with eigenvalue $(\lambda, \mu) \in S^1 \times S^1$ is a continuous function $G : Y^{(m)} \to S^1$ satisfying

$$G(Hz) = \lambda G(z), \qquad G(Vz) = \mu G(z).$$

Let E_H denote the group of vertical eigenvalues that admit an H-invariant continuous eigenfunction. Thus

$$\langle e(\theta/m) \rangle \subseteq E_H(Y^{(m)}), \tag{4.6}$$

and equality in (4.6) requires the return argument discussed next.

5. The full spectrum as a proof program

This section records the proposed spectrum calculation in a form that makes each missing input visible. Theorems 3.4, 3.5, and 4.3 remain independent of its conditional propositions.

5.1 Controlled returns and the vertical upper bound

The needed dynamical statement is the following.

Controlled-return lemma, open. There is a point $z \in Y^{(m)}$ such that, for every integer sequence q_ℓ with

$$q_\ell \theta/m \longrightarrow 0 \pmod 1, \tag{5.1}$$

there are integers a_ℓ satisfying

$$H^{a_\ell} V^{q_\ell} z \longrightarrow z. \tag{5.2}$$

A proof must specify a nonsingular continuity point in Siefken's parameter space, show that the sheet coordinate returns for every sequence satisfying (5.1), and obtain the horizontal corrections a_ℓ within the same continuity fiber. The joint convergence in (5.2) requires more than density of horizontal orbits.

Proposition 5.1, conditional vertical spectrum. If the controlled-return lemma holds, then

$$E_H(Y^{(m)}) = \langle e(\theta/m) \rangle. \tag{5.3}$$

Proof under the stated hypothesis. Inclusion from right to left follows from Theorem 4.3. Let $\mu \in E_H(Y^{(m)})$, with eigenfunction G. For the returns in (5.2),

$$G(H^{a_\ell} V^{q_\ell} z) = \mu^{q_\ell} G(z).$$

Continuity and (5.2) imply

$$\mu^{q_\ell} \longrightarrow 1$$

for every sequence satisfying (5.1).

Since θ/m is irrational, the rule

$$\chi(q\theta/m) = \mu^q$$

defines a homomorphism on the dense cyclic subgroup $(\theta/m)\mathbb{Z}/\mathbb{Z}$ of the circle. The preceding sequential criterion makes χ continuous at zero. It extends uniquely to a continuous character of the circle, hence has the form $\chi(t) = e(kt)$ for some $k \in \mathbb{Z}$. Therefore $\mu = e(k\theta/m)$. □

5.2 Horizontal eigenvalues

The intended joint spectrum is

$$E(Y^{(m)}) = \{(1, e(k\theta/m)) : k \in \mathbb{Z}\}. \tag{5.4}$$

Proposition 5.1 would determine the vertical coordinate of (5.4). Excluding nontrivial horizontal eigenvalues requires another lemma.

Let $\mathbb{S}_6$ denote the 6-adic solenoid. Its Pontryagin dual is $\mathbb{Z}[1/6]$.

Solenoidal continuity-fiber lemma, open. There is an uncountable set A of irrational row parameters such that, for each $\alpha \in A$:

1. the Siefken coding is continuous at a point over α;

2. the closure of its horizontal orbit carries the rotation by α on $\mathbb{S}_6$;

3. restricting a continuous horizontal eigenfunction with eigenvalue λ to that orbit produces a continuous character of $\mathbb{S}_6$.

The third item must include the return-continuity argument that extends the character from the dense cyclic orbit to the full solenoid.

Proposition 5.2, conditional horizontal triviality. If the solenoidal continuity-fiber lemma holds, then every continuous horizontal eigenvalue on $Y^{(m)}$ is 1.

Proof under the stated hypothesis. A continuous character of $\mathbb{S}_6$ is indexed by some $q \in \mathbb{Z}[1/6]$. Restriction to the orbit generated by α gives

$$\lambda = e(q\alpha). \tag{5.5}$$

The value of q may depend on α. If $\lambda \neq 1$, then for each fixed nonzero q, equation (5.5) has a countable set of real solutions α. The union over the countable group $\mathbb{Z}[1/6] \setminus \{0\}$ is countable, contrary to the uncountability of A. The case $q = 0$ gives $\lambda = 1$. $\square$

Together, the two open lemmas would prove (5.4).

5.3 The finite phase extension

The geometric compiler uses four honeycomb phases. Let Ξ_m denote the corresponding finite extension of $Y^{(m)}$. The proposed discrete eigenvalue group is

$$E(\Xi_m) = \left\{ \left((-1)^{\varepsilon}, (-1)^{\delta} e(k\theta/m) \right) : \varepsilon, \delta \in \{0,1\},\ k \in \mathbb{Z} \right\}. \tag{5.6}$$

Formula (5.6) follows immediately if the extension is conjugate to the product with $(\mathbb{Z}/2\mathbb{Z})^2$, where H and V increment the two phase coordinates. The actual collared phase system has yet to be identified explicitly with that product. Alternatively, a proof can describe its finite cocycle and calculate the characters of that cocycle. Until one of these routes is completed, (5.6) remains conditional.

5.4 The proposed suspension module

The affine face-center lattice of the geometric realization has basis

$$a = (-3, 3/2), \qquad b = (0, 3),$$

with dual basis

$$a^* = (-1/3, 0), \qquad b^* = (1/6, 1/3).$$

Assuming the joint spectrum (5.4), the phase formula (5.6), an explicit topological conjugacy between the symbolic suspension and the fixed-frame geometric presentation, and absence of additional continuous eigenvalues introduced by collaring, the fixed-frame suspension has eigenmodule

$$M_m = \frac{1}{2}\mathbb{Z}\, a^* + \left(\frac{1}{2}\mathbb{Z} + \frac{\theta}{m}\mathbb{Z} \right) b^*. \tag{5.7}$$

The displayed generators describe the intended geometric eigenvectors only after both the phase product and the symbolic-suspension/geometric conjugacy are established. Equality with the full continuous eigenmodule also requires the two spectrum upper bounds and preservation of the complete continuous spectrum under that conjugacy.

6. Relative eigenfields on rotating hulls

6.1 Definition

Let X be an $\mathbb{R}^2$-tiling system. A relative eigenfield is a pair

$$\kappa : X \longrightarrow (\mathbb{R}^2)^*, \qquad f : X \longrightarrow S^1,$$

such that κ is continuous and translation-invariant and

$$f(\mathcal{T} - t) = e\big(\kappa(\mathcal{T}) \cdot t\big) f(\mathcal{T}). \tag{6.1}$$

The wavevector may follow an intrinsic frame of the tiling. This makes (6.1) suitable for a hull containing every global rotation and reflection.

6.2 Extension from a fixed frame

Suppose $X_0 \subseteq X$ is a fixed-frame subsystem and there is a continuous, translation-invariant map

$$R : X \longrightarrow O(2)$$

such that every $\mathcal{T} \in X$ has a unique representation

$$\mathcal{T} = R(\mathcal{T})\mathcal{T}_0, \qquad \mathcal{T}_0 \in X_0. \tag{6.2}$$

The uniqueness assumption holds for a geometric presentation whose tile shapes determine a common orthogonal frame and have no frame-changing symmetries. Verifying these hypotheses for the intended geometric realization is a separate obligation, not a consequence of the abstract statement.

Proposition 6.1, induced relative eigenfield. If $f_k : X_0 \to S^1$ is a continuous eigenfunction with wavevector $k \in (\mathbb{R}^2)^*$, then

$$\begin{aligned} \kappa_k(\mathcal{T}) &= R(\mathcal{T})k, \\ \widehat{f_k}(\mathcal{T}) &= f_k\big(R(\mathcal{T})^{-1}\mathcal{T}\big) \end{aligned} \tag{6.3}$$

defines a relative eigenfield on X.

Proof. Continuity and well-definedness follow from (6.2). Write $\mathcal{T} = Q\mathcal{T}_0$, with $Q = R(\mathcal{T})$. Translation preserves the frame, and

$$\mathcal{T} - t = Q(\mathcal{T}_0 - Q^{-1}t).$$

Therefore

$$\begin{aligned} \widehat{f_k}(\mathcal{T} - t) &= f_k(\mathcal{T}_0 - Q^{-1}t) \\ &= e(k \cdot Q^{-1}t) f_k(\mathcal{T}_0) \\ &= e((Qk) \cdot t) \widehat{f_k}(\mathcal{T}), \end{aligned}$$

which is (6.1). □

Proposition 6.1 constructs relative eigenfields from known fixed-frame eigenvectors. A classification of every relative eigenfield would require a separate completeness theorem.

7. Conditional nonconjugacy of rotating hulls

Let Ω_m be the full rotating geometric hull presenting the four-phase extension of $K^{(m)}$. Let $X_m \subseteq \Omega_m$ be the closed rotating saturation of the fixed-frame presentation of $Y^{(m)}$. Thus X_m is the component hull studied in this section, whereas Ω_m also contains presentations of all other Kari-Culik components.

The component-hull argument requires:

1. X_m has an intrinsic orientation map satisfying (6.2).
2. Its fixed-frame continuous eigenmodule is exactly M_m from (5.7).

Both hypotheses remain open for the intended geometric realization.

An A-conjugacy, for $A \in O(2)$, is a homeomorphism $\Phi : X_m \to X_n$ satisfying

$$\Phi(\mathcal{T} - t) = \Phi(\mathcal{T}) - At. \tag{7.1}$$

The case $A = I$ is an ordinary translation conjugacy.

Theorem 7.1, conditional component-hull separation. Under hypotheses 1 and 2, an A-conjugacy $X_m \to X_n$ can exist only when $m = n$.

Proof under the stated hypotheses. Restrict Φ to a fixed-frame minimal component of X_m. The map

$$\mathcal{T} \longmapsto R_n(\Phi(\mathcal{T}))$$

is continuous and translation-invariant. Minimality makes it a constant orthogonal matrix Q.

Pulling the target relative eigenfield induced by a fixed-frame wavevector k through Φ gives the source wavevector

$$A^T Q k.$$

The inverse conjugacy gives the reverse correspondence. Hence an orthogonal matrix $O = A^T Q$ satisfies

$$OM_n = M_m. \tag{7.2}$$

The closure of (5.7) is

$$\overline{M_m} = \bigcup_{r \in \frac{1}{2}\mathbb{Z}} (ra^* + \mathbb{R}b^*)\,. \tag{7.3}$$

Its connected components are parallel lines with the unique direction $\mathbb{R}b^*$. Equation (7.2) and orthogonality give

$$Ob^* = \pm b^*.$$

Intersecting (7.2) with $\mathbb{R}b^*$ yields

$$\frac{1}{2}\mathbb{Z} + \frac{\theta}{n}\mathbb{Z} = \frac{1}{2}\mathbb{Z} + \frac{\theta}{m}\mathbb{Z}. \tag{7.4}$$

The inclusion from left to right in (7.4) gives integers r, k with

$$\frac{\theta}{n} = \frac{r}{2} + \frac{k\theta}{m}.$$

Irrationality of θ forces $r = 0$ and $m = kn$, so $n \mid m$. The reverse inclusion gives $m \mid n$. Therefore $m = n$. $\square$

The algebra after (7.2) is complete. Establishing (7.2) for the intended component hulls depends on the intrinsic-frame and full-eigenmodule hypotheses.

To deduce the same conclusion for an A-conjugacy $\Omega_m \to \Omega_n$, one needs the additional open hypothesis that the chosen rotating Siefken-component hull is intrinsically recognizable and is therefore carried from X_m to X_n. Theorem 7.1 by itself makes no claim about the full hulls.

8. Open proof obligations

The following tasks delimit the unfinished part of the paper.

1. **Locate the multiplier claim precisely.** State the exact theorem from Siefken that identifies the Sturmian factor, its parameter space, and the subsystem on which the coding is continuous. Avoid extending this claim to every component of the Kari-Culik hull without proof.

2. **Prove controlled returns.** Select a nonsingular parameter point and prove (5.2) for every sequence satisfying (5.1). The proof must coordinate the vertical return, the winding-sheet return, the horizontal correction, and continuity of the symbolic coding.

3. **Prove the solenoidal continuity-fiber lemma.** Identify the relevant 6-adic fibers, exhibit uncountably many suitable irrational parameters, and justify extension of the character from each dense horizontal orbit.

4. **Describe the four-phase system.** Give an explicit conjugacy to a $(\mathbb{Z}/2\mathbb{Z})^2$ product extension or compute the finite cocycle directly. This is required before applying Fourier decomposition.

5. **Identify the geometric presentation.** Construct the explicit conjugacy from the symbolic suspension to the geometric hull. Prove that the intended tile shapes provide the global continuous frame and unique decomposition required by (6.2).

6. **Prove fixed-frame spectral completeness.** Show that (5.7) contains every continuous eigenvector of the collared symbolic suspension and that the geometric conjugacy preserves this full module.

7. **Control minimal components under conjugacy.** Prove that the lifted Siefken component is intrinsically recognizable, or prove the required module statement on every minimal component to which a conjugacy might send it.

8. **Classify relative eigenfields if needed.** Proposition 6.1 constructs a family of relative eigenfields. A claim that these exhaust all relative eigenfields needs an additional theorem.

Until obligations 2 through 7 are discharged, the full eigenmodule formula, component-hull separation, and full-hull separation must remain propositions under explicit hypotheses.

9. Discussion

The winding extension is a genuine cyclic skew product. The sheet at one row determines every sheet, and the cocycle records division rows modulo m. It preserves aperiodicity and entropy and supplies a candidate refinement of the vertical equicontinuous behavior on the lifted Sturmian subsystem.

The explicit function F_m detects part of that change. Its eigenvalue $e(\theta/m)$ is established there conditional on Subsystem input 4.1. Showing that the equicontinuous behavior actually differs with m requires an upper bound on the full continuous spectrum. A list of constructed eigenvalues cannot provide that upper bound.

Relative eigenfields offer the appropriate language for geometric hulls containing all orientations. Their abstract construction is elementary once an intrinsic frame and a fixed-frame eigenfunction are available. The remaining difficulty lies in spectral completeness and in the behavior of minimal components under conjugacy.

References

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